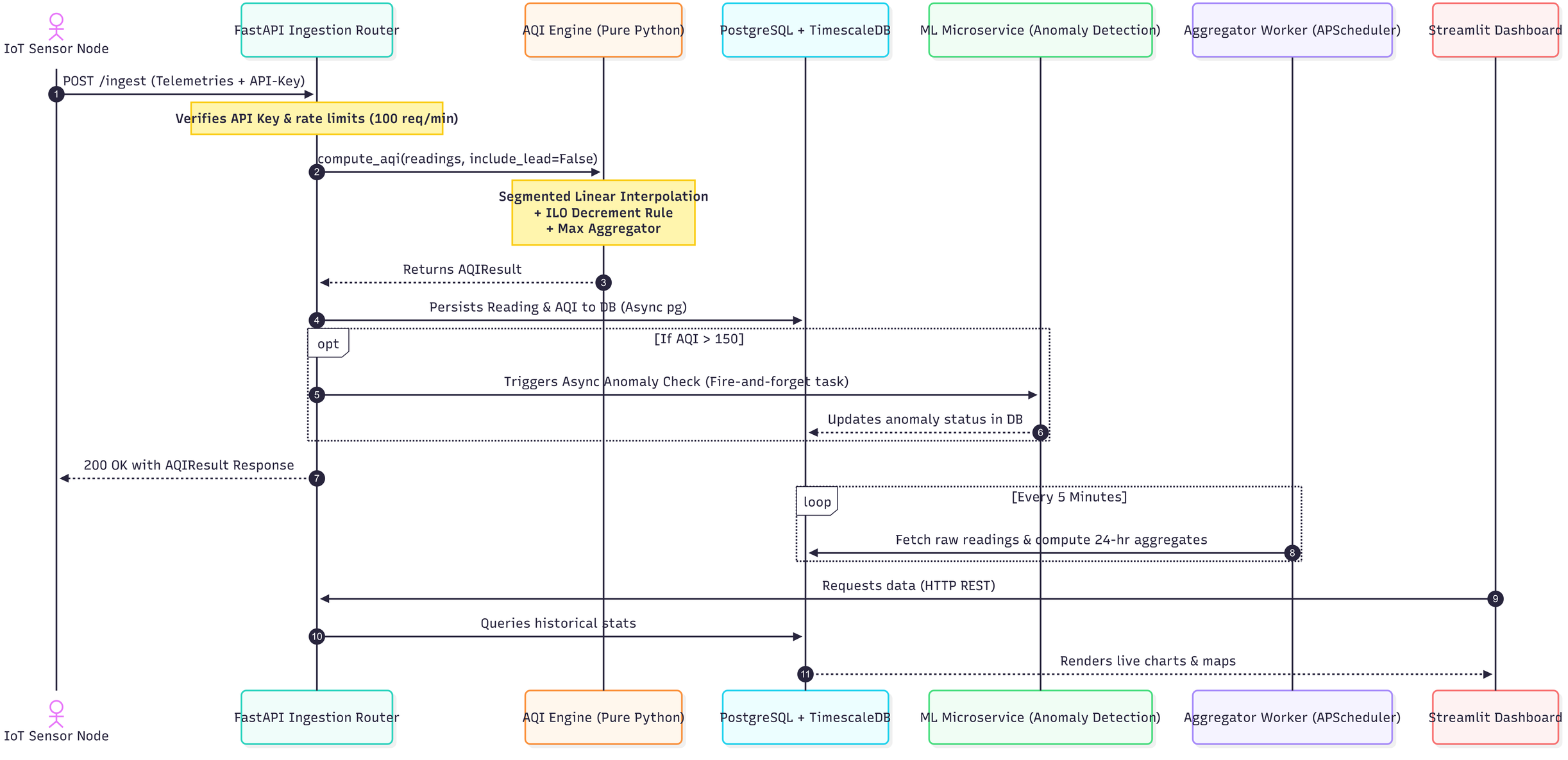

IoT Sensor Node
FastAPI Ingestion Router
AQI Engine (Pure Python)
PostgreSQL + TimescaleDB
ML Microservice (Anomaly Detection)
Aggregator Worker (APScheduler)
Streamlit Dashboard
POST /ingest (Telemetries + API-Key)
1
Verifies API Key & rate limits (100 req/min)
compute_aqi(readings, include_lead=False)
2
Segmented Linear Interpolation
+ ILO Decrement Rule
+ Max Aggregator
Returns AQIResult
3
Persists Reading & AQI to DB (Async pg)
4
opt
[If AQI > 150]
Triggers Async Anomaly Check (Fire-and-forget task)
5
Updates anomaly status in DB
6
200 OK with AQIResult Response
7
loop
[Every 5 Minutes]
Fetch raw readings & compute 24-hr aggregates
8
Requests data (HTTP REST)
9
Queries historical stats
10
Renders live charts & maps
11
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